

**ATTACHMENT 13 – RESPONSE LETTER FROM MODERN CONTINENTAL
COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED
DECEMBER 2, 1999
(7 pages)**



Letter No. C09B2-0155

ROUTING REQUEST				
99119 - C09B2				
A		DISTRIB.	W	WO
✓		RE	✓	
		ARE		
		OE	✓	
		AOE		
		LFE	✓	
R		FE	✓	
		C09B2-Sch	✓	
		C09B2 C.C.	✓	
		Proj Eng	✓	
FILE		CN - 9 09B2 - 01		
FILE		CN - 9 09B2 - 49-01-001		
FILE				



DEFICIENCY REPORT

CONTRACT NO: 99119-C09B2

CONTRACT NAME: I-90 SEAPORT ACCESS TUNNEL FINISHES

DATE OF DEFICIENCY: December 1, 1999

DEFICIENCY REPORT NO: 001

DATE ISSUED: December 1, 1999

FIELD ENGINEER: P. C. Aikele

TO: Bob Coutts	DISCUSSED WITH: Bill Britto & Jim Martin
SPECIFICATION SECTIONS: 723.480 - 2.08	SUBMITTAL NO: N/A
SCHED. ACTIVITY NO: N/A	PAY ITEM NO: N/A
REQUIREMENT: The ceiling adhesive anchors shall be installed according to the project specifications 723.480-2.08 and manufacturer's requirements so that when the ceiling system is attached to them they do not fail, i.e. pull out.	
DEFICIENCY: At least five ceiling adhesive anchors in the area of ceiling module HOV 079 have failed, that is pull-out has been observed, since the hanging of the ceiling module. See the attached sketch for the locations of the failed anchors and the amount of pull-out that has been observed.	
RECOMMENDED ACTION: (1) MCC shall contact the adhesive anchor manufacturer to arrange for a site visit and investigation to determine the reason(s) for the failure of the anchors and the proposed corrective actions. (2) MCC shall submit the proposed corrective actions to the Engineer for approval prior to implementation.	RESPONSE DUE DATE: 12/08/99 [Signature] 12-2-99 RESIDENT ENGINEER DATE
CORRECTIVE ACTION TAKEN: PLEASE SEE ATTACHED RESPONSE (ACTIONS) BY MCC & LETTER FROM POWERS FASTENERS (SUBMITTAL No 723.480.04-A)	FORECAST COMPLETION DATE: [Signature] 12-6-99 CONTRACTOR QUALITY CONTROL MANAGER OR APPROPRIATE CONTRACTOR REPRESENTATIVE DATE
DESIGN CHANGE: YES NO	PROJECT ENGINEER DATE
CORRECTIVE ACTION VERIFIED:	ACTUAL COMPLETION DATE: [Signature] DATE RESIDENT ENGINEER



CENTRAL ARTERY/TUNNEL

Computation SheetSubject HOV MOCK-UP CEILING ANCHOR
BOLT PULL OUT MEASUREMENTS

Page _____ of _____

Made by JIM BARRETTDate 10/08/99

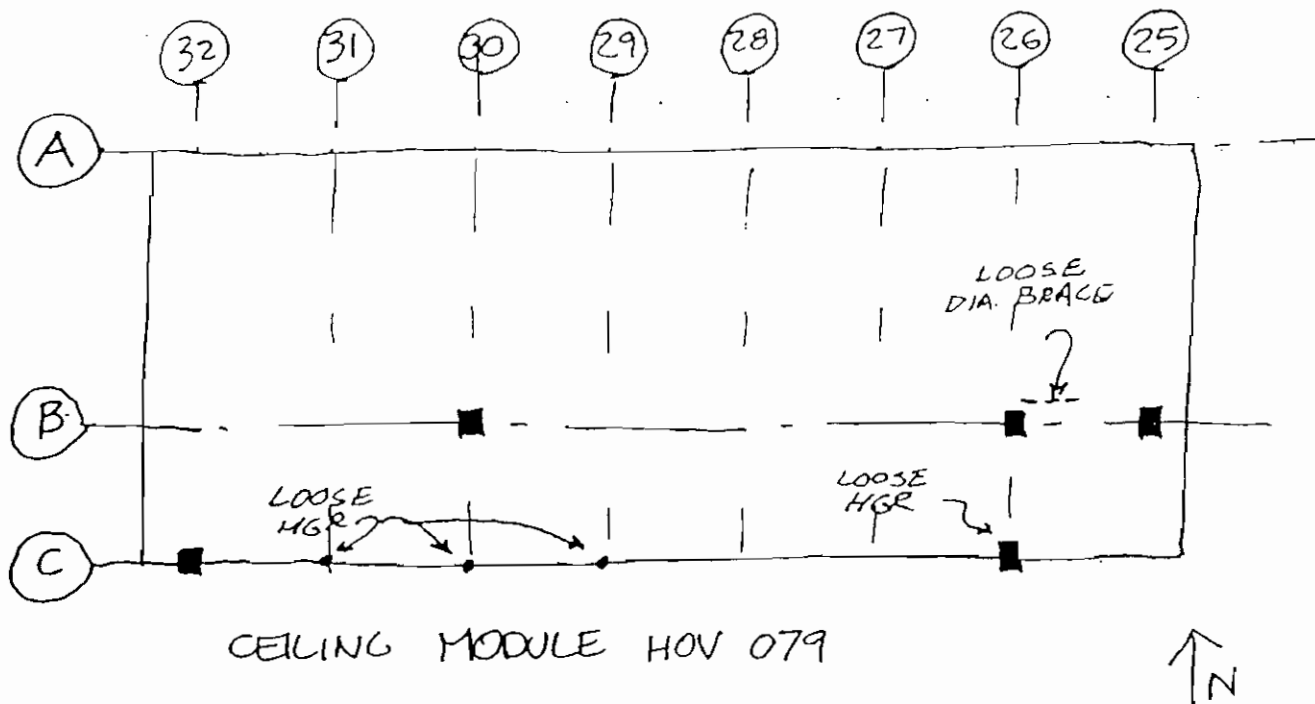
Checked by _____

Date _____

PLATE LOCATION	CORNER:	GAP BETWEEN TUNNEL ROOF AND CEILING HANGER PLATE			
		NE	NW	SE	SW

B-25	0	0	9/16"	9/16"
B-26 *	0	3/8"	0	3/8"
B-30	1/8"	3/16"	0	1/16"
C-26	1/4"	5/16"	0	0
C-32	0	0	3/16"	1/8"

* 4-BOLT (CROSS-BRACING) PL



DR-1 – Adhesive Anchor Failures

In response to the above mentioned Deficiency Report the following corrective actions were / will be taken by MCC upon approval by B/PB.

- 1) On October 21, 1999 a Powers / Rawl Rep. was on site to investigate and determine to reason(s) for the failure of the anchors. Based on information gathered on site, which included a visual inspection of the anchors in question, a determination of failure could not be made.
- 2) MCC submitted a letter / procedure from Powers Fasteners dated Nov. 30, 1999 to B/PB for approval (submittal no 723.480-016-A, also attached). Upon approval MCC will schedule a time to perform and complete this procedure.

Powers. **FASTENERS**

Tuesday, November 30, 1999

The Modern Continental Construction Co., Inc.
6 Necco Court
Boston MA 02210

Attn: Mr. Neil Joyce
Ph: (617) 695-2277
Fax: (617) 695-2272

Powers Fasteners, Inc.
2 Powers Square
New Rochelle, NY
USA 10801

Telephone: 1-(914) 235-6300
Fax: 1-(914) 576-6483

Web Site: www.powers.com
E-Mail: williamson@powers.com

Re: CO9B2 I-90 Seaport Access

Dear Mr. Joyce:

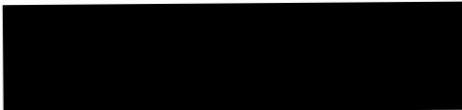
Pursuant to your request, please accept the following suggestions for the repair and replacement of the adhesive anchors where movement was observed on the HOV ceiling mock up.

- The adhesive anchor rods in question should be removed from the base material with care, preventing damage to the concrete base material and preserving the integrity of the original anchor location. This can be done with the use of a hydraulic hollow core ram placed directly on the surface to minimize spalling.
- To install replacement adhesive anchors in the same anchor hole, it is recommended to remove any debris from the hole. Upon removal of the threaded rod, the anchor holes should be blown clean with compressed air, brushed clean with a nylon brush, and blown clean again, to ensure that freshly injected adhesive has a solid substrate to bond to.
- The threaded rod removed from the anchor hole should be replaced with a new, clean threaded rod to ensure a good mechanical interlock between the freshly injected adhesive and threads of the anchor rod.
- Adhesive anchor installation procedures, as were previously approved by the Project Engineer and in accordance with Powers Fasteners' published Installation Guidelines should be performed.

- Adhesive anchors should be allowed to fully cure. Proof loading per job site specifications should be performed to verify the repair. Upon successful passing of the proof load test, the fixture can be attached to the threaded anchor rod by applying a torque to the nut and washer that does not exceed the published maximum torque range (50 – 90 foot-pounds).

Should you have any questions or require further information, please feel free to contact me at (914) 235-6300. Thank you for your continued interest in Powers Fasteners.

Very truly yours,
Powers Fasteners, Inc.



Ray Williamson
Engineering Manager

cc: William H. Alderman, P.E. – Powers Fasteners, Inc.
Jack Armour – Powers Fasteners, New England
Rich Tamulionis – Newman Renner Colony